Glaze Search · Quick Start

For new users: install once, then follow one example click by click, from a photo of a pot to the glaze recipes that could make it, and to the pots in the database that look like it. Every screenshot below is the real app doing exactly these steps (6 November 2026); if you do the same, you see the same.

1 · Install once (or use it online)

No install: open `https://huggingface.co/spaces/FBIWarninggg/glaze_search` and log in to Hugging Face (more free GPU time). **On your computer:** clone the branch, then run five launchers in order (Mac: the `.command` files, Apple Silicon M1 or newer).

```
git clone -b nghia_mike_ceramic_search --single-branch https://github.com/haohaozai/GC2-Ensemble-Model-Pottery.git
```

#	WINDOWS	MACOS	WHAT IT DOES	WHEN
1	`SETUP.bat`	`SETUP.command`	Python environment and the pretrained models (about 3.7 GB); needs `uv` once	once
2	`DOWNLOAD_DATA_ORIGINAL_AND_SEGMENTATION` .bat / .command		photos, cut-outs and search index (8 archives, about 6.15 GB); run again if it stops	once
3	`CHECK_DATA.bat`	`CHECK_DATA.command`	every line should say OK	once
4	`INSTALL_GLAZE_MODEL.bat`	`INSTALL_GLAZE_MODEL.command`	the recipe models from the team's `test_similarity.zip` (drag it onto the .bat; Mac: put it in Downloads)	once
5	`RUN_GLAZE_SEARCH.bat`	`RUN_GLAZE_SEARCH.command`	starts the app at `http://127.0.0.1:8770`; keep the window open	every time

2 · The example, click by click

The task: you have a photo of a glossy blue bowl (held in your hand) and fire to cone 6 in an electric kiln (oxidation). Which Glazy recipe makes this glaze?

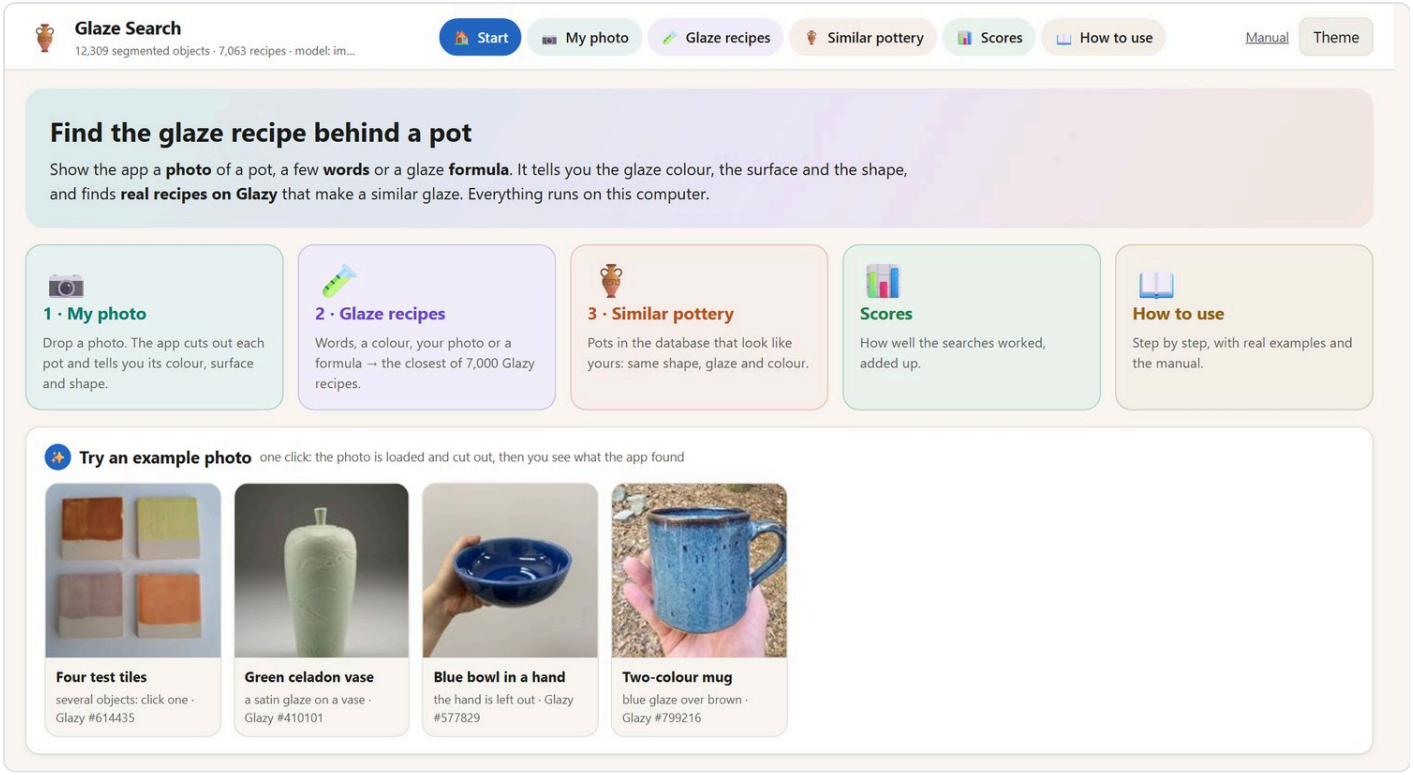
1 Start page: load the photo

DO

Under **Try an example photo** click **Blue bowl in a hand**. (With your own pot: click 📷 **My photo** and drop, paste or browse the photo.)

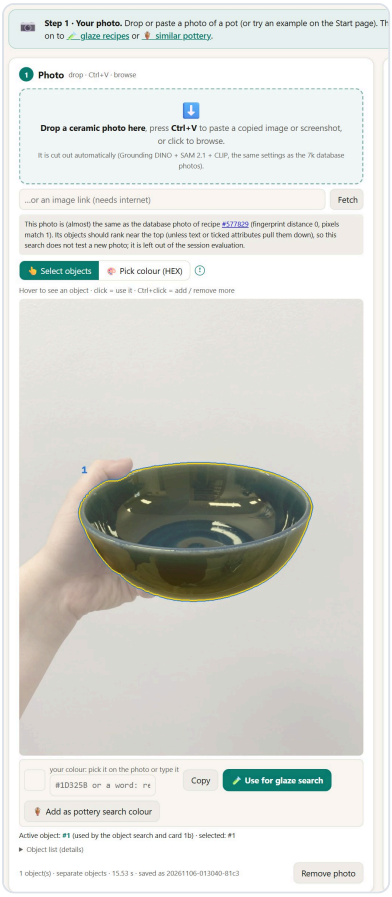
THEN

The app opens 📷 **My photo** and cuts the pot out: a few seconds with an NVIDIA GPU, about a minute on a CPU or a Mac.



The 🏠 Start page. The third example photo is the one used here.

2 My photo: the pot is found and cut out



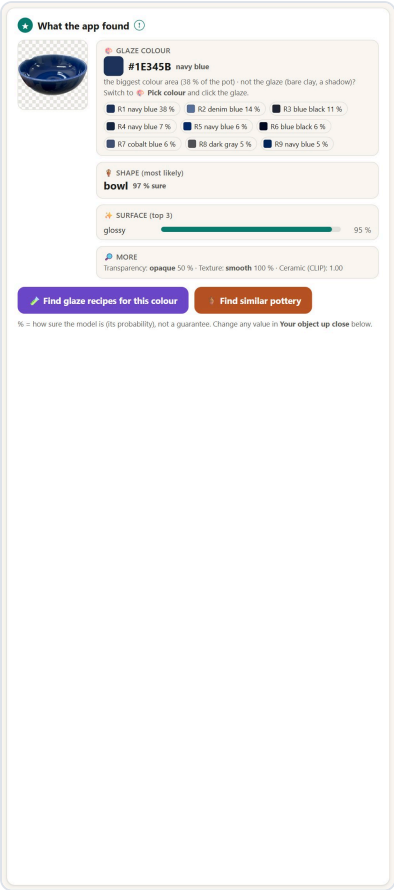
📷 My photo after the example loaded (the photo half of the page).

You see: the bowl outlined and shaded (the shading marks the object in use); your hand is **not** part of the object.

The grey note says this photo is already in the Glazy database (recipe #577829). That is expected for an example photo: it is left out of the session scores.

Diagram: A1 Grounding DINO finds the bowl · A2 SAM 2.1 traces its outline · A3 CLIP checks it is ceramic

3 Read “What the app found”



What the app found, on the right of the photo page.

PANEL	SHOWS
🎨 Glaze colour	**#1E345B navy blue**: the biggest colour area (38 % of the pot), then the other colour regions
🏺 Shape	**bowl**, 97 % sure
✨ Surface	**glossy**, 95 %
🔎 More	opaque 50 % · smooth 100 % · ceramic (CLIP) 1.00

Check the colour is the **glaze**, not bare clay or a shadow. If not: 🎨 **Pick colour** and click on the glaze.

Diagram: A4 colours (ColorThief + CIELAB) · A5 six attributes

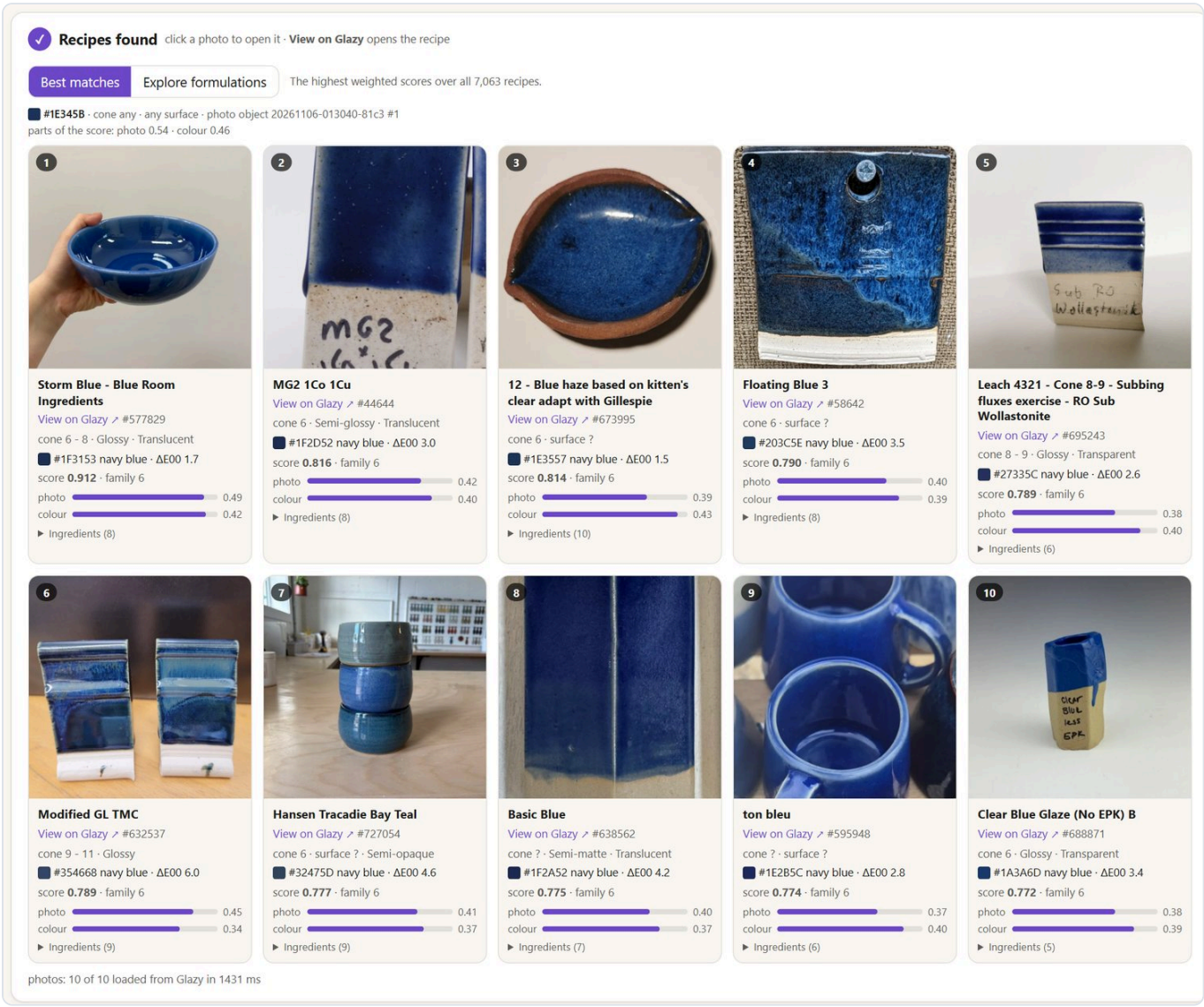
4 Press 🧪 Find glaze recipes for this colour

DO

Press the purple button at the bottom of **What the app found**.

YOU SEE

The 🧪 **Glaze recipes** page, already searched with your photo and its colour: *photo 0.54 · colour 0.46*.



The first result: #1 Storm Blue - Blue Room Ingredients (score 0.912), #2 MG2 1Co 1Cu (0.816), #3 12 - Blue haze based on kitten's clear adapt with Gillespie (0.814).

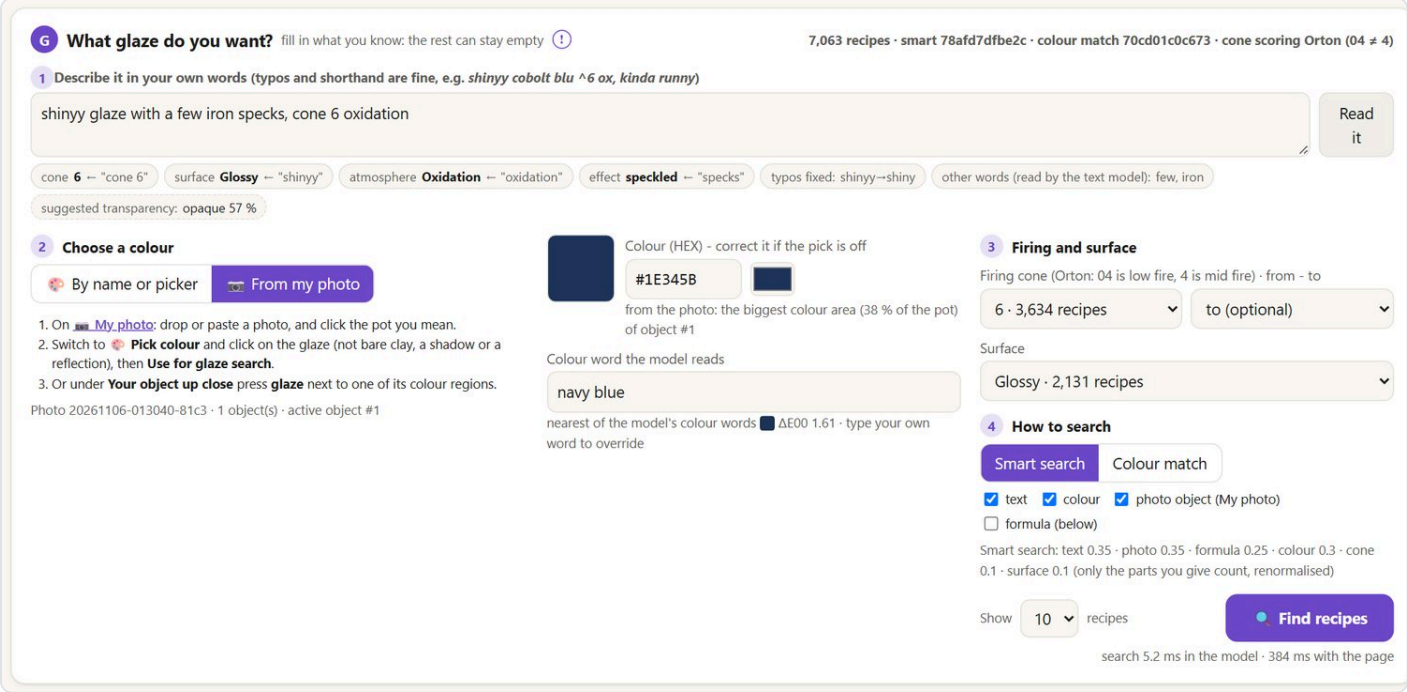
5 Describe what you want, in your own words

DO

Scroll up to **1 · Describe it**, type `shinyy glaze with a few iron specks, cone 6 oxidation` and press **Read it**. Typos are fine.

YOU SEE

Chips with what it understood: **cone 6** ← "cone 6" · **Glossy** ← "shinyy" · **Oxidation** · effect **speckled** ← "specks" · typos fixed: shinyy → shiny. Cone and surface are filled in under 3.

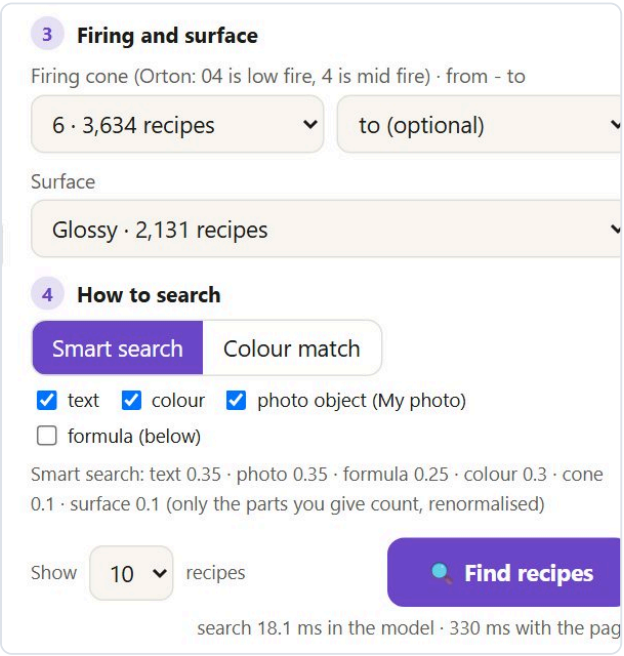


Card G after Read it: the chips under the sentence; the colour is still the photo's #1E345B ("from the photo: the biggest colour area").

> **Do not name the colour in the words when you use a photo.** If the sentence says a colour (for example "dark blue"), Read it puts *that* colour into Colour (HEX) and the photo's colour is no longer used. If it happens, type the photo's HEX (here `#1E345B`) back into Colour (HEX).

Diagram: B3 read the words (rules + fuzzy matching) · B2 MiniLM reads the whole sentence

6 Check firing and engine, then search



3 · Firing and surface and 4 · How to search.

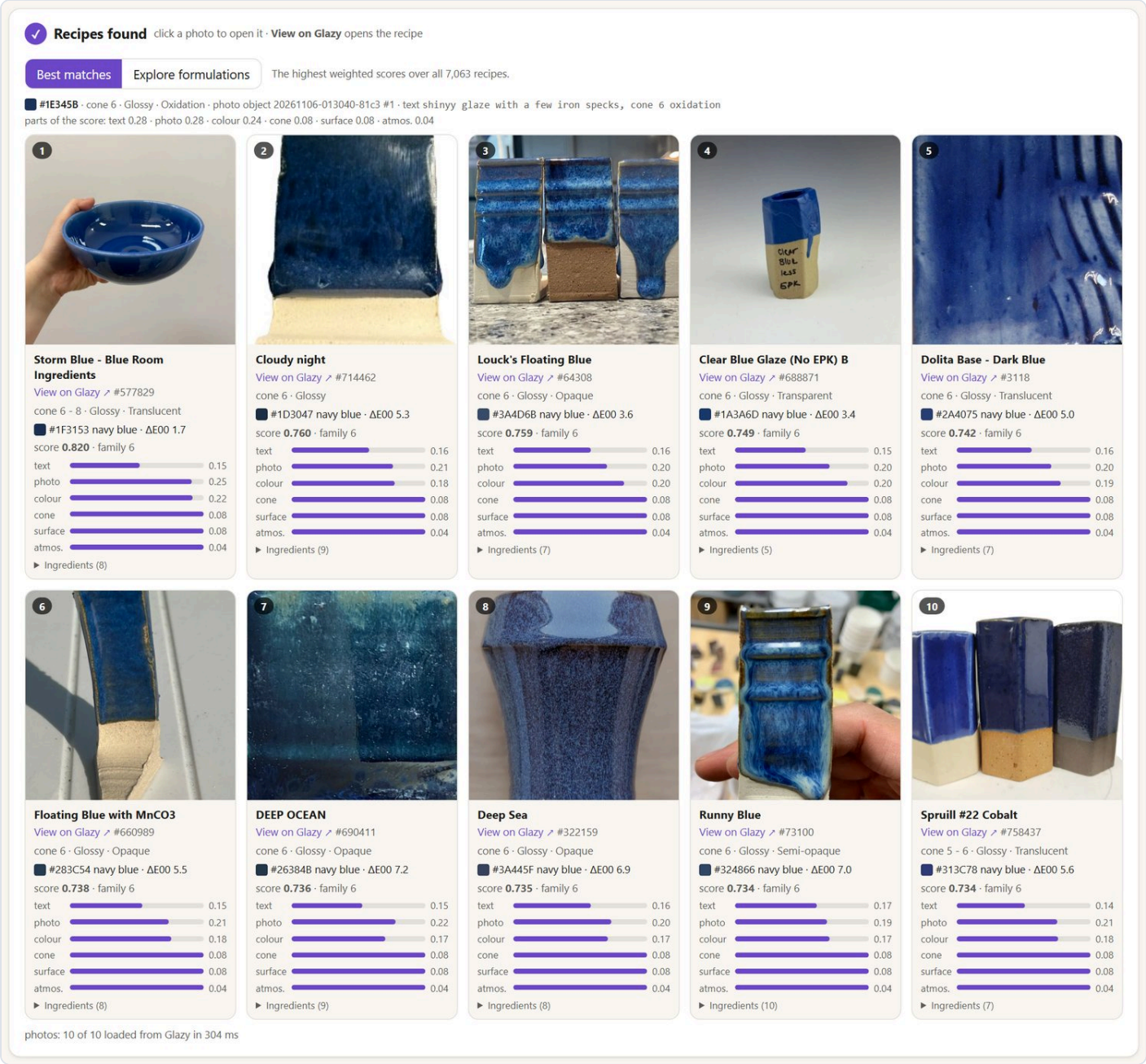
3 · Firing and surface: cone 6 and Glossy, from your words (cone **04** is low fire, not cone 4).

4 · How to search: Smart search, with text, colour and photo object ticked. **Show 10**.

Do: press 🔍 **Find recipes**. It takes under a second.

Diagram: join → C2 smart search (3 trained heads on frozen MiniLM and DINOv2) → rank every recipe

7 The result: Best matches

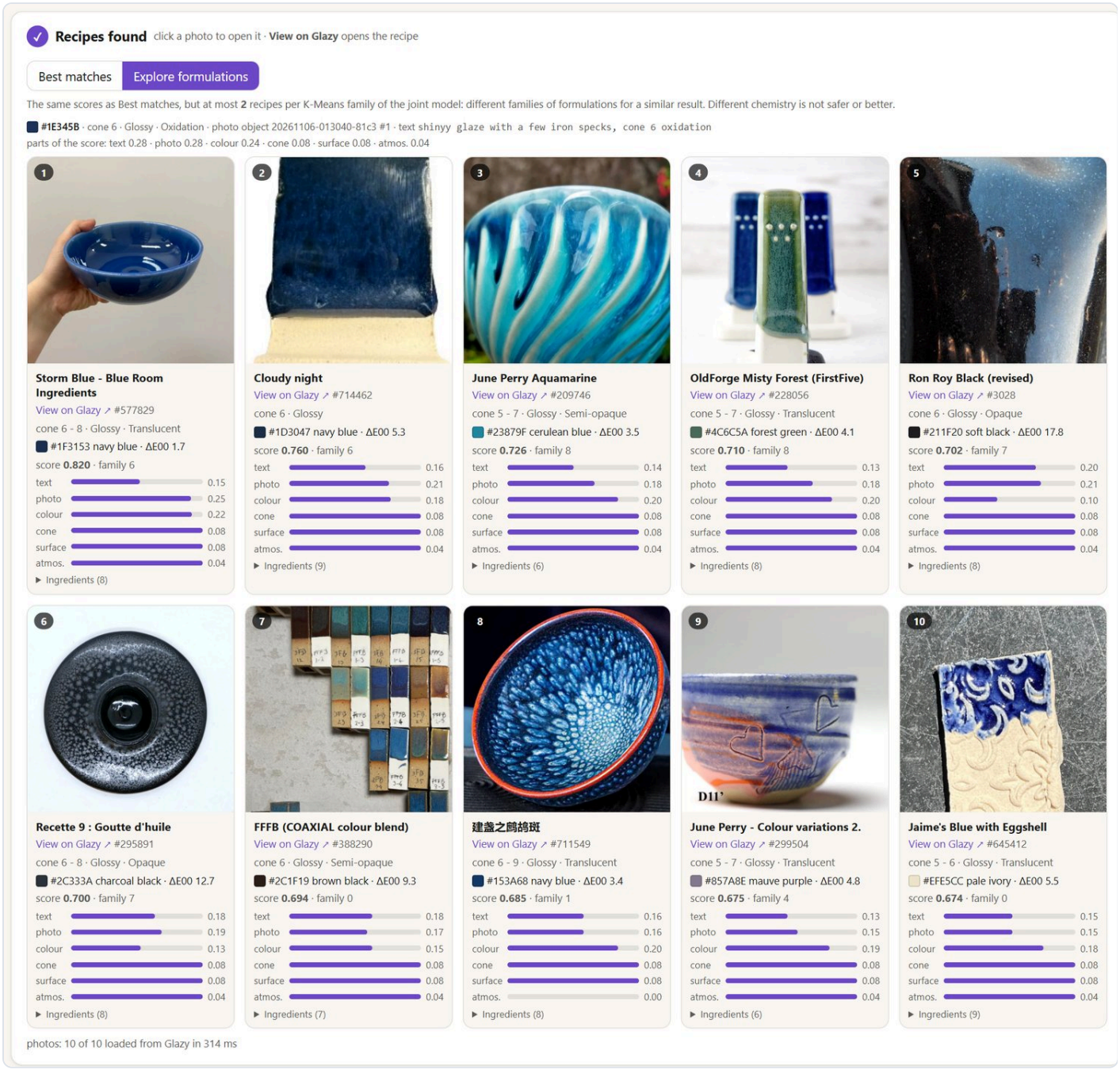


Best matches, scored over all 7,063 recipes with *text 0.28 · photo 0.28 · colour 0.24 · cone 0.08 · surface 0.08 · atmos. 0.04*. Every photo comes from Glazy.

> **How to read a card:** the Glazy photo and name (**View on Glazy ↗** opens the recipe); cone, surface and transparency; the colour with its **ΔE00** (below about 5 the colours look alike, above 10 they are different colours); the score, with bars for what each part added.

8 Explore formulations

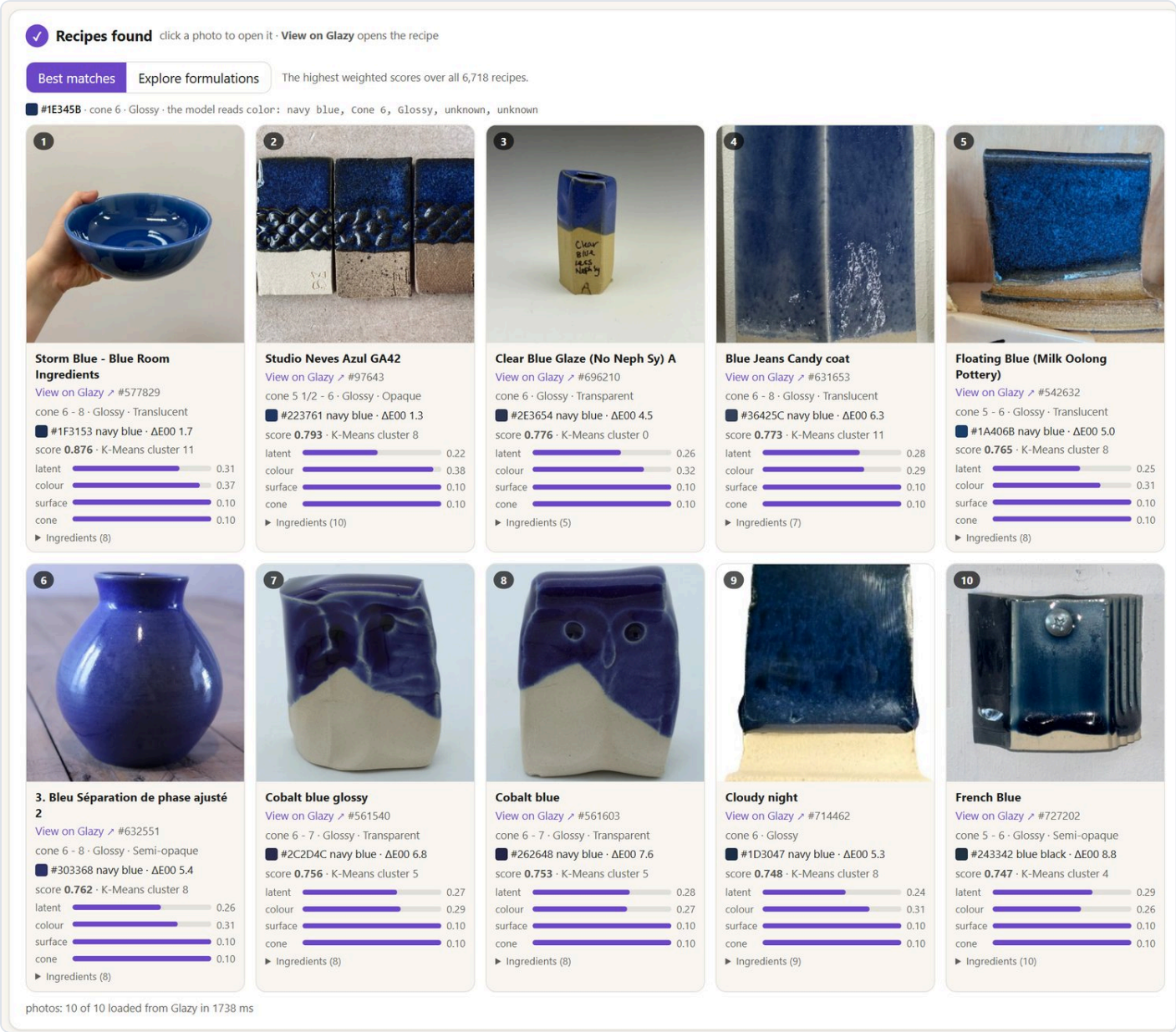
Click the **Explore formulations** tab: the same scores, but at most 2 recipes per chemistry family, so you see different bases for a similar look.



Explore formulations: Storm Blue - Blue Room Ingredients · Cloudy night · June Perry Aquamarine · OldForge Misty Forest (FirstFive) · Ron Roy Black (revised) …

9 (Optional) Ask the second engine: Colour match

Click **Best matches**, switch **4 · How to search** to **Colour match** and press 🔍 **Find recipes**. It reads only colour + cone + surface (`color: navy blue, Cone 6, Glossy, unknown, unknown`) and compares them with each recipe's chemistry: 0.40 latent (learned) match + 0.40 colour + 0.10 surface + 0.10 cone.



Colour match: #1 Storm Blue - Blue Room Ingredients (0.876), #2 Studio Neves Azul GA42 (0.793), #3 Clear Blue Glaze (No Neph Sy) A (0.776).

10 Decide

Open the top cards with **View on Glazy ↗** for the ingredients. Prefer a small ΔE00 and the right cone for your kiln, then **test-fire a small tile** before you mix a big batch.

The answer: Storm Blue - Blue Room Ingredients (Glazy #577829, #1F3153 navy blue · ΔE00 1.7), first in Smart search (score 0.820) and in Colour match (0.876).
It is this bowl's own recipe: the example photo comes from the Glazy database, so the app found the exact recipe from the photo and a sentence with a typo. With your own pot there is no “exact” recipe: the list gives the closest glazes (here Cloudy night, Louck's Floating Blue …), and you choose with ΔE00 and the cone.

3 · The same photo: pots that look like it (🏺 Similar pottery)

Glaze recipes finds **recipes** for a colour. Similar pottery finds **pots**: it compares your object with 12,309 objects cut out of Glazy photos by shape, glaze look, surface and colour regions. It uses no words and no formula unless you add them.

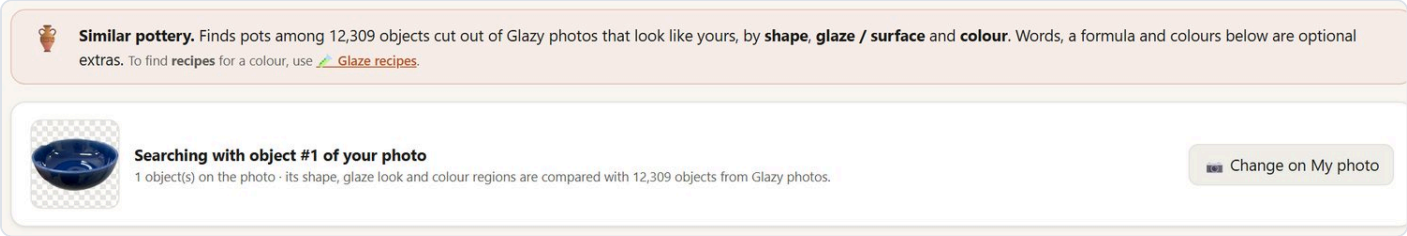
11 Press 🏺 Find similar pottery

DO

Go back to 📷 **My photo** (the bowl is still loaded) and press the second button under **What the app found**: 🏺 **Find similar pottery**.

YOU SEE

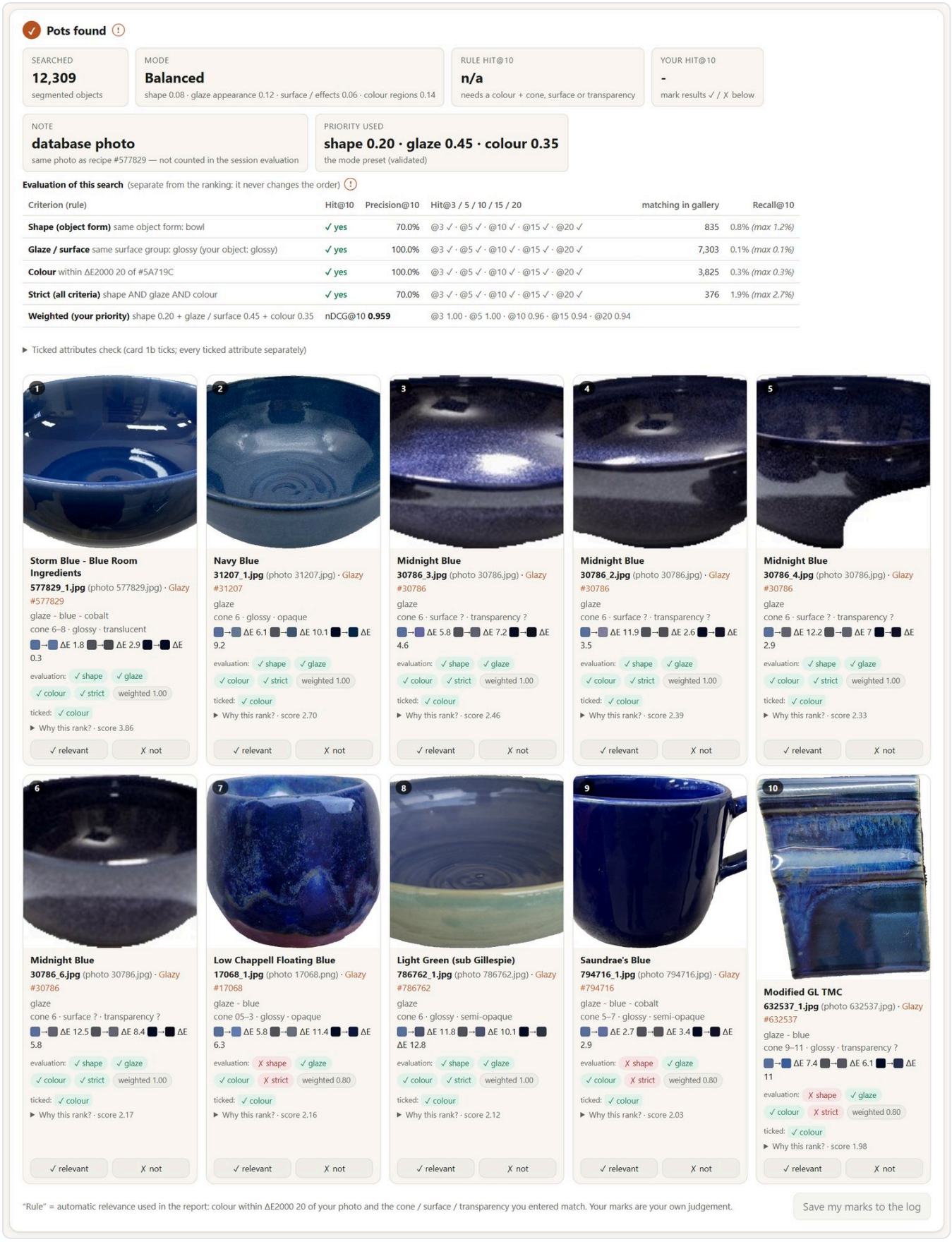
The 🏺 **Similar pottery** page, already searched with object #1 of your photo in **Balanced** mode (under 2 s).



Top of the Similar pottery page: the object used for the search.

Diagram: DINOv2 (glaze look, silhouette) · shape measures · surface / effect classifiers · colour regions (ΔE2000) → each score standardised → weighted sum

12 Read “Pots found”



Pots found (Balanced): #1 Storm Blue - Blue Room Ingredients (score 3.86), #2 Navy Blue (2.70), #3 Midnight Blue (2.46).

13 Check the evaluation, then read the cards

Every search is checked automatically, under the stat boxes. The check never changes the order of the results.

CRITERION (RULE)	HIT@10	PRECISION@10
Shape: same object form (bowl)	✓ yes	70.0%
Glaze / surface: same surface group (glossy)	✓ yes	100.0%
Colour: within ΔE2000 20 of the pot's colour	✓ yes	100.0%
Strict: shape AND glaze AND colour	✓ yes	70.0%
Weighted by your priority (shape 0.20 · glaze 0.45 · colour 0.35)	nDCG@10 **0.959** (1 = perfect order)	

Each card shows the cut-out object (click it for the whole Glazy photo), the recipe name and **Glazy #** link, cone · surface · transparency, the colour pairs *your region → its region* with their ΔE, the evaluation chips (✓ shape ✓ glaze ✓ colour ✓ strict), and **Why this rank?** with the similarity of each part. #1 is the bowl's own photo: the grey note "database photo" says so, and such a search is not counted in the session scores.

Optional: mark pots ✓ **relevant** or ✗ **not**; "Your Hit@10" is then worked out from your marks, and **Save my marks to the log** stores them for the report.

14 Try the presets: what matters most to you?

Under **5 · Search**, click a preset. The page searches again at once (under a second).

PRESET	PRIORITY (SHAPE · GLAZE · COLOUR)	TOP 3 (SCORE)
Balanced	0.20 · 0.45 · 0.35	Storm Blue - Blue Room Ingredients (3.86) · Navy Blue (2.70) · Midnight Blue (2.46)
Shape	0.65 · 0.20 · 0.15	Storm Blue - Blue Room Ingredients (3.70) · Navy Blue (3.03) · Midnight Blue (2.84)
Glaze / surface	0.05 · 0.85 · 0.10	Storm Blue - Blue Room Ingredients (5.59) · Navy Blue (3.18) · Janice Hill's Teal from Chrome and Cobalt (2.85)
Colour	0.05 · 0.15 · 0.80	Storm Blue - Blue Room Ingredients (3.17) · Navy Blue (2.46) · [44] K3 Koper Kobalt Tin (2.38)

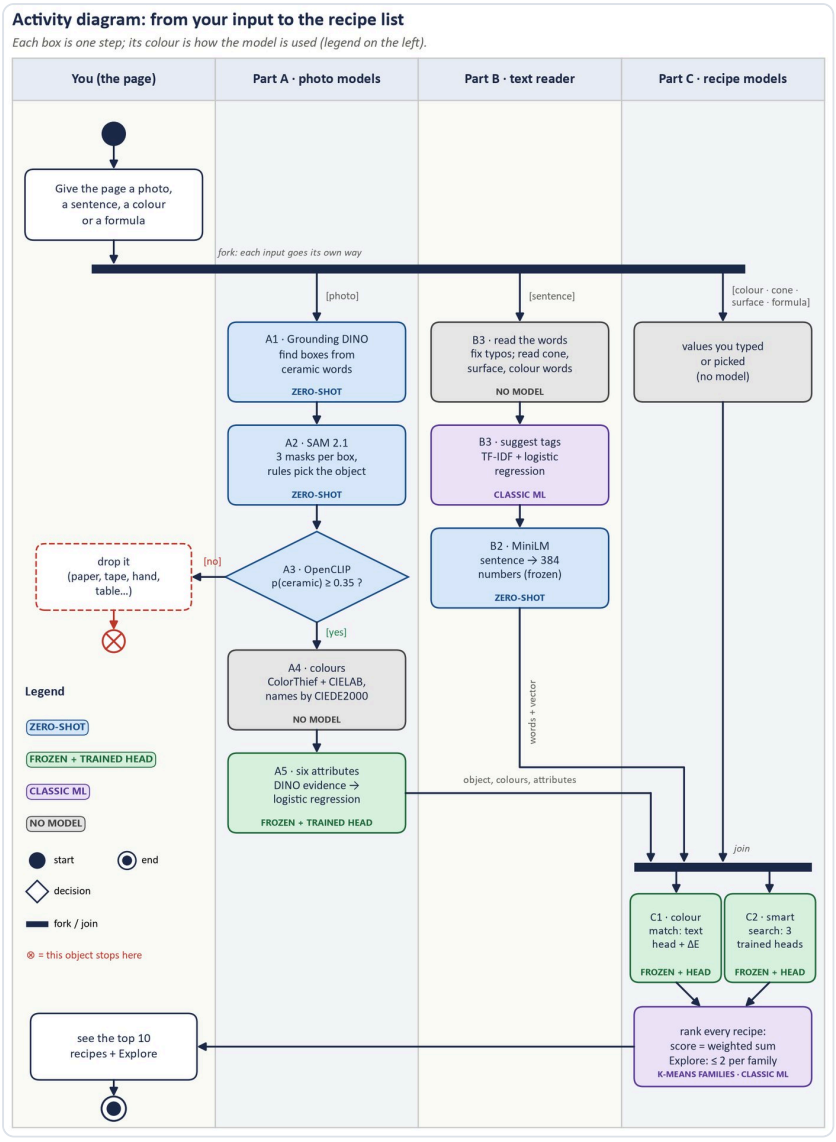
> **With a photo, leave card 2 (Words about the pot) empty.** We tried adding cone 6 and Glossy: the words then weigh 0.35 against 0.40 for the whole photo, and many recipes share the same few words, so pots of other colours came up. The bowl's own recipe fell from #1 to **#21** and nDCG@10 from 0.959 to 0.608. Use the words when you have **no** photo. (The *Hard filters* in card 2 only work with *Recipes (older model)*, not with *Objects*.)

4 · Your results at a glance

STEP	SEARCH	TOP 3 (SCORE)
4	photo + colour	Storm Blue - Blue Room Ingredients (0.912) · MG2 1Co 1Cu (0.816) · 12 - Blue haze based on kitten's clear adapt with Gillespie (0.814)
7	Smart search: words + photo + colour + cone + surface + atmosphere	Storm Blue - Blue Room Ingredients (0.820) · Cloudy night (0.760) · Louck's Floating Blue (0.759)
9	Colour match: colour + cone + surface	Storm Blue - Blue Room Ingredients (0.876) · Studio Neves Azul GA42 (0.793) · Clear Blue Glaze (No Neph Sy) A (0.776)
11–12	Similar pottery, Balanced: the photo object only	Storm Blue - Blue Room Ingredients (3.86) · Navy Blue (2.70) · Midnight Blue (2.46)

5 · What happened behind each step

STEP	BOX IN THE ACTIVITY DIAGRAM	HOW THE MODEL IS USED	BUILT FROM
2	A1 Grounding DINO · A2 SAM 2.1 · A3 CLIP	pretrained, zero-shot: a text prompt, a box prompt, 30 sentences	segmentation pipeline
3	A4 colours · A5 six attributes	ColorThief + CIELAB (no model) · DINO evidence → small trained layer	segmentation pipeline
5	B3 read the words · B2 MiniLM	rules + fuzzy matching · frozen text model	glaze search (own)
6–7	join → C2 smart search → rank	3 small trained heads on frozen MiniLM and DINOv2	own weights, trained with the **method of** `training.ipynb` in test_similarity.zip (shared 256-d space, masked InfoNCE, composition groups); recipe text from the text branch's tag cleaning
8	Explore: ≤ 2 per K-Means family	K-Means families of the joint model	the Explore rule of `retrieval.ipynb`
9	C1 colour match	frozen MiniLM + 2 trained layers (text ↔ chemistry)	**test_similarity.zip, unchanged**: catalog, vectors and heads; `retrieve_glazes` of `retrieval.ipynb`, same results
11–14	object search (Similar pottery)	DINOv2 glaze look + silhouette, shape measures, surface classifiers, colour regions	multimodal object model; the words card uses the text branch



The activity diagram of one search.

6 · Problems

IF…	THEN…
the colour is the bare clay or a shadow	📷 My photo → 🎨 Pick colour and click on the glaze
the colour changed after Read it	your words named a colour: type the photo's HEX back into Colour (HEX), or leave the colour out of the words
the wrong pot is used	click the pot you mean on 📷 My photo
Similar pottery shows pots of other colours	empty card 2 (Words about the pot) when you search with a photo, or choose the **Colour** preset
online: “Hugging Face's free GPU: RuntimeError”	try again in a minute; the Space owner can restart the Space
online: “free GPU limit”	log in to Hugging Face, or wait: the allowance renews

More: the Manual (all six pages, six worked examples), Process and models, Pretrained models, Evaluation, in `docs/manuals`.

Glaze Search Quick Start · screenshots and numbers from the app on 6 November 2026 (Glaze Search server with the installed models: colour match 70cd01c0c673, smart search 78afd7dfbe2c). Recipe photos and names belong to their authors on glazy.org.